

Beom Park

Apollo 11 Postdoctoral Fellow

School of Aeronautics and Astronautics

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Research Interests

Dynamical systems theory; Multi-body and perturbed dynamical environments; Nonlinear and chaotic orbital/attitude dynamics; Spacecraft trajectory design and optimization; Guidance, navigation, and control

Employment

- 2025.8- **Apollo 11 Postdoctoral Fellow**, Purdue University, U.S.
Advisor: Professor Kathleen C. Howell
- 2025.5- **Postdoctoral Researcher**, Korea Advanced Institute of Science and Technology
2025.7 (KAIST), Korea
Advisor: Professor Jaemyung Ahn

Education

- 2021-2025 **Ph.D.** in Aeronautics and Astronautics, Purdue University, U.S.
Advisor: Professor Kathleen C. Howell
Thesis Title: A Frequency-Based Model Hierarchy within Cislunar Space [[link](#)]
Committee: Professor Carolin Frueh, Professor Kenshiro Oguri, Dr. Diane Davis
- 2018-2021 **M.S.** in Aeronautics and Astronautics, Purdue University, U.S.
Advisor: Professor Kathleen C. Howell
Thesis Title: Low-Thrust Trajectory Design for Tours of the Martian Moons [[link](#)]
Committee: Professor Carolin Frueh, Professor James Longuski
- 2012-2018 **B.S.**, College of Liberal Studies (Aerospace Engineering), Seoul National University, Republic of Korea

Honors and Awards

- 2025-2027 *Apollo 11 Postdoctoral Fellowship* [[link](#)]:
Awarded by School of Aeronautics and Astronautics, Purdue University
- 2025 *Magoon Research Excellence Award* [[link](#)]:
Awarded by College of Engineering, Purdue University
- 2025 *John V. Breakwell Student Award* [[link](#)]
Awarded by American Astronautical Society

- 2024 *Korea Aerospace Research Institute's Travel Grant:*
Awarded for participating in the International Astronautical Congress 2024, Milan, Italy
- 2021-2025 *Kwanjeong Scholarship:*
Awarded by Kwanjeong Educational Foundation, Korea, for Ph.D. studies abroad
- 2018-2020 *Kwanjeong Scholarship:*
Awarded by Kwanjeong Educational Foundation, Korea, for M.S. studies abroad
- 2018 *Graduation with Honor:*
Awarded by the Seoul National University Alumni Association for the academic excellence upon graduation

Research Projects

- 2024-2025 Lunar satellite constellation design leveraging frozen orbits [\[link\]](#)
Sponsor: Intuitive Machines
Relevant publications: [J1](#). [C2](#).
- 2022-2025 Model hierarchy within cislunar space
Partial sponsor: NASA Johnson Space Center
Relevant publications: [J2](#). [J3](#). [J4](#). [J5](#). [J6](#). [C1](#). [C3](#). [C4](#). [C5](#). [C6](#). [C8](#). [P1](#). [P2](#).
- 2021-2024 Trajectory design and analysis for the IM-1 & Khon-1 missions [\[link\]](#)
Sponsor: Intuitive Machines
Relevant publication: [C7](#).
- 2021 Low-thrust trajectory design in multi-body regimes
Partial sponsor: NASA Goddard Space Flight Center, Lunar IceCube (LIC, [\[link\]](#))
Relevant publication: [J7](#). [C9](#). [C10](#).

Publications

Journal Papers

- J1. [Park, B.](#), [Howell, K.](#), and [Stewart, S.](#) (2025) "A Frequency-Domain Differential Corrector for Quasi-Periodic Trajectory Design and Analysis," *Physica D: Nonlinear Phenomena*, 482, 134882. [\[link\]](#)
- J2. [Park, B.](#), [Sanaga, R. R.](#), and [Howell, K.](#) (2025) "Numerical Assessment of Intermediate Models in a Frequency-Based Hierarchy for the Cislunar Domain," *Journal of Guidance, Control, and Dynamics*, In Press.
- J3. [Park, B.](#), [Sanaga, R. R.](#), and [Howell, K.](#) (2025) "Bridging Ephemeris Transition Gaps: Quasi-Periodic Extensions for the Hill Restricted Four-Body Problem in Cislunar Space," *Acta Astronautica*, 233, 168-182. [\[link\]](#)
- J4. [Park, B.](#), [Sanaga, R. R.](#), and [Howell, K.](#) (2025) "A Frequency-Based Hierarchy of Dynamical Models in Cislunar Space: Leveraging Periodically and Quasi-Periodically Perturbed Models," *Celestial Mechanics and Dynamical Astronomy*, 137, 5. [\[link\]](#) - invited to a webinar [\[link\]](#)

- J5. Park, B., and Howell, K. (2025) “Characterization of Earth-Moon L2 Halo Analogs in an Ephemeris Model Utilizing the Elliptic Restricted Three-Body Problem,” *Advances in Space Research*, 75, 6. [\[link\]](#)
- J6. Park, B., and Howell, K. (2024) “Assessment of Dynamical Models for Transitioning from the Circular Restricted Three-Body Problem to an Ephemeris Model with Applications,” *Celestial Mechanics and Dynamical Astronomy*, 136, 6. [\[link\]](#) - invited to a webinar [\[link\]](#)
- J7. Canales, D., Gupta, M., Park, B., and Howell, K. (2022) “A Transfer Trajectory Framework for the Exploration of Phobos and Deimos Leveraging Resonant Orbits,” *Acta Astronautica*, 194, 263-276. [\[link\]](#)

Conference Proceedings / National (International) Presentations

- C1. Sanaga, R. R., Park, B., and Howell, K. C. (2025) “A Unified Numerical Transition Scheme between Dynamical Models within Cislunar Space,” *35th AAS/AIAA Space Flight Mechanics Meeting*, Kaua’i, Hawaii, U.S., January 19-23, 2025. [\[link\]](#)
- C2. Park, B., Howell, K. C., and Stewart, S. (2025) “Elliptical Lunar Frozen Orbit Constellation Design within a Model of Evolving Fidelity,” *35th AAS/AIAA Space Flight Mechanics Meeting*, Kaua’i, Hawaii, U.S., January 19-23, 2025. [\[link\]](#)
- C3. Park, B., Sanaga, R. R., and Howell, K. C. (2025) “Numerical Assessment of Intermediate Models in a Frequency-Based Hierarchy for the Cislunar Domain,” *35th AAS/AIAA Space Flight Mechanics Meeting*, Kaua’i, Hawaii, U.S., January 19-23, 2025. [\[link\]](#) - awarded John V. Breakwell Student Award
- C4. Park, B., Sanaga, R. R., and Howell, K. C. (2024) “Bridging Ephemeris Transition Gaps: Quasi-Periodic Extensions for the Hill Restricted Four-Body Problem in Cislunar Space,” *75th International Astronautical Congress*, Milan, Italy, October 14-18, 2024. [\[link\]](#) - awarded Korea Aerospace Research Institute’s Travel Grant
- C5. Park, B. and Howell, K. C. (2024) “Characterizing Transition-Challenging Regions Leveraging the Elliptic Restricted Three-Body Problem: L2 Halo Orbits,” *AIAA SciTech Forum*, Orlando, Florida, U.S., January 8-12, 2024. [\[link\]](#)
- C6. Park, B. and Howell, K. C. (2023) “Leveraging the Elliptic Restricted Three-Body Problem for Characterization of Multi-Year Earth-Moon L2 Halos in an Ephemeris Model,” *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, Montana, U.S., August 13-17, 2023. [\[link\]](#)
- C7. Hoffman, A., Park, B., and Howell, K. C. (2022) “Trajectory Design for a Secondary Payload Within a Complex Gravitational Environment: The Khon-1 Spacecraft,” *AAS/AIAA Astrodynamics Specialist Conference*, Charlotte, North Carolina, U.S., August 7-10, 2022. [\[link\]](#)
- C8. Park, B. and Howell, K. C. (2022) “Leveraging Intermediate Dynamical Models for Transitioning from the Circular Restricted Three-Body Problem to an Ephemeris Model,” *AAS/AIAA Astrodynamics Specialist Conference*, Charlotte, North Carolina, U.S., August 7-10, 2022. [\[link\]](#)
- C9. Park, B., Howell, K. C., and Folta, D. C. (2021) “Design of Low-Thrust Transfers from an NRHO to Low Lunar Orbits: Applications for Small Spacecraft,” *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, Montana (Virtual), U.S., August 9-11, 2021. [\[link\]](#)

- C10. Canales, D., Gupta, M., Park, B., and Howell, K. C. (2021) “Exploration of Deimos and Phobos Leveraging Resonant Orbits,” *31st AAS/AIAA Spaceflight Mechanics Meeting*, Charlotte, North Carolina (Virtual), U.S., February 1-3, 2021. [[link](#)]

Preprints (Under Review)

- P1. Sanaga, R. R., Park, B., and Howell, K. (2025) “A Unified Numerical Transition Scheme between Dynamical Models within Cislunar Space,” *The Journal of Astronautical Sciences*, Under Review.
- P2. Sanaga, R. R., Park, B., and Howell, K. (2025) “Unified Continuation Scheme for Periodic and Quasi-Periodic Structures across Various Models in the Cislunar Domain,” *Communications in Nonlinear Science and Numerical Simulation*, Under Review. [[link](#)]

Talks

- 2025 Invited talks in Korea at (1) Sejong University, (2) Seoul National University, (3) Yonsei University and (4) Korea Astronomy and Space Science Institute on “Frequency-Based Lunar GNSS Constellation Design and Analysis” ([J1.](#))
- 2025 Invited talk at *Celestial Mechanics and Dynamical Astronomy*’s webinar on [J4](#).[\[link\]](#), selected “as being particularly innovative and/or having had significant recent impact”
- 2024 Invited talk at *Celestial Mechanics and Dynamical Astronomy*’s webinar on [J6](#).[\[link\]](#), selected “as being particularly innovative and/or having had significant recent impact”

Teaching Experiences

- 2020 **Teaching Assistant**, School of Aeronautics and Astronautics, Purdue University.
- AAE 440/590 (Spacecraft Attitude Dynamics)
- AAE 532 (Orbit Dynamics)
- 2019 **Teaching Assistant**, Department of Mathematics, Purdue University.
- MA 161 (Plane Analytic Geometry And Calculus I)

Professional Experience

- 2013-2015 **Air Surveillance Operator (Sergeant)**, Republic of Korea Air Force

Academic Services

Journal of Spacecraft and Rockets (2 papers), The Journal of Astronautical Sciences (4), Astrodynamics (1)